A Theory of Economic Slack

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Terminology

Before we start, we define the core terms used throughout the book. Some of the terms are not used commonly in economics, some are often misconstrued, and some are used in the book in a way that is atypical. Either way, it is useful to clarify the meaning of these terms upfront.

Slack terms

- The term *slack* describes situations in which goods or services are available for sale but trades do not occur. Sometimes the goods are already produced; sometimes they are available to be produced at negligible marginal cost. In both cases, trades that would generate value for sellers and buyers do not materialize.
- The *slack rate*, denoted by u, is the share of goods and services that remain unsold.
- The *efficient slack rate*, denoted by u^*, is the slack rate that maximizes social welfare by minimizing the social costs from slack—both unused resources and resources devoted to matching.
- The *slack gap*, denoted by $u - u^*$, is the gap between the actual slack rate and the efficient slack rate. It measures departure from social efficiency.
- The term *slackish* describes a class of markets where slack is the central mechanism that equalizes supply and demand. Slackish does not mean the market is always too slack; rather, it refers to the structural mechanism through which the market operates. It means that market tightness is the key mediating variable, whether the market is inefficiently slack or inefficiently tight.

Model terms

- The term *equilibrium* is used to denote time-invariant points of differential equations or dynamical systems (instead of *steady state*). An equilibrium point has $\dot{x}(t) = 0$. This follows usage in dynamical systems and conveys the idea of a balanced state, which may be reached quickly.
- The term *solution* is used to denote the allocation that satisfies all the equations of a model (instead of *equilibrium*). This describes more transparently what the allocation is.
- The *matching function* refers to the aggregate function that mediates trades in slackish markets. With s sellers and b buyers looking to trade, the matching function $m(s, b)$ gives the number of trades that take place. It summarizes the random, costly, imperfect process of finding trading partners without modeling that process in detail.
- The term *tightness* refers to the central equilibrating variable in slackish markets. Tightness is denoted by θ and defined as the ratio of the two arguments in the matching function: the ratio b/s between the number of buyers and the number of sellers. Since trading probabilities are determined by tightness, tightness summarizes the state of the market.
- The *matching elasticity* refers to the elasticity of the matching function with respect to the number of sellers, holding the number of buyers fixed. The matching elasticity is a structural property of the matching function, denoted by η, and generally endogenous to tightness.
- The *matching wedge* denotes the gap between goods purchased and goods consumed caused by matching costs. Buyers must devote part of their expenditure to matching costs, so the amount spent on goods and services that are actually consumed is less than the total amount spent. If the wedge is τ, consuming one unit requires purchasing $1 + \tau$ units. Because matching probabilities depend on tightness, the wedge is endogenous to tightness. In the labor market, where matching costs take the form of recruiting costs, we refer to the matching wedge as the *recruiting wedge*.
- The *market supply* denotes the amount of goods actually sold given market tightness and market price. It is the effective supply; the *market capacity* is the notional market supply: the amount of goods that sellers place on the market and would like to sell given market tightness and market price. Because each good sells only with probability less than one, effective supply falls short of notional supply.
- The *market demand* denotes the amount of goods that buyers actually purchase given market price and market tightness. That is the effective demand. The notional market demand is the amount that buyers would like to consume. Because consuming one

good requires purchasing more than one, effective demand exceeds notional demand. Buyers purchase goods from sellers through *visits*. Visits are the demand-side input into the market's matching function, just as goods for sale are the supply-side input.

Labor market terms

- The term *unemployed workers* describes workers who are available and willing to work but unable to find a job. This definition is consistent with that used by the US Bureau of Labor Statistics (BLS). Based on the definition, the concept of voluntary unemployment is an oxymoron. Being unemployed means wanting a job but not finding one: it is by definition involuntary. Someone who does not have a job and does not want one (which is presumably what is meant with voluntary unemployment) is *out of the labor force*—not unemployed. Unemployed workers represent slack on the labor market.
- The *unemployment rate* is the number of unemployed workers as a share of the labor force.
- The term *job vacancies* denotes positions that firms report as open and for which they are recruiting. This definition is consistent with that used by the BLS. Vacancies measure recruiting effort; they are not a measure of unmet labor demand. Some vacancies never result in hiring, because the job-filling probability is always less than 1. Job vacancies correspond to visits on the labor market.
- The *vacancy rate* is the number of job vacancies as a share of the labor force.
- The term *full employment* describes a labor market that operates efficiently. Accordingly, the full-employment rate of unemployment (FERU) is the socially efficient rate of unemployment. Full employment therefore does not mean zero unemployment and does not automatically imply stable prices.
- The *unemployment gap* denotes the distance of labor market slack from social efficiency. It is computed by comparing the actual unemployment rate to the FERU.

Product market terms

- The term *idle capacity* describes the productive capacity of a firm that is available to be sold but cannot find customers and remains unsold. Idle capacity represents slack on the product market.
- The *idleness rate* is the share of productive capacity that remains idle.
- The term *idle labor* describes workers employed by firms but unproductive because their services are not bought by customers.

Curves

- The *Beveridge curve* is the negative relationship between the slack rate and the visit rate. In the labor market specifically, it is the negative relationship between the unemployment rate and the vacancy rate. In dynamic slackish models, it appears once market flows are balanced and the slack rate reaches equilibrium.
- The *Okun curve* is the relationship between the logarithm of output and slack. The slopes of Okun curves are useful to identify the shocks that drive business cycles. *Okun's law* is the empirical observation that the labor Okun curve is downward sloping.
- The *labor Okun curve* (which is the standard Okun curve) is the relationship between log output and unemployment (labor-market slack).
- The *product Okun curve* is the relationship between log output and idle capacity (product-market slack).
- The term *Phillips curve* denotes the negative relationship between inflation and slack. The slackish Phillips curve links inflation to the slack gap, unlike the New Keynesian Phillips curve, which links inflation to an output gap or marginal costs.

Policy terms

- The *inflation-slack coincidence* denotes the property that inflation is on target whenever the slack rate is efficient (instead of *divine coincidence*).
- The *product-labor coincidence* denotes the property that product-market slack is efficient whenever labor-market slack is efficient.
- The term *sufficient statistics* refers to statistics used in formulas describing efficient allocations and optimal policies. They have two defining properties, which distinguish them from structural parameters. First, they should be valid across a broad range of models. Second, they should be estimable directly from data.

Math terms

- The terms *positive* and *negative* are used for $x \geq 0$ and $x \leq 0$ (instead of *nonnegative* and *nonpositive*). The terms *strictly positive* and *strictly negative* are used for $x > 0$ and $x < 0$. This follows French usage and avoids the cognitive load from "non-" constructions.
- The terms *increasing* and *decreasing* are used for weakly monotone functions (instead of *nondecreasing* and *nonincreasing*). The terms *strictly increasing* and *strictly decreasing* are used for strictly monotone functions. This again avoids the cognitive load from "non-" constructions.